

Secure Password Manager User Guide

Introduction

Secure Password Manager is a free, open-source, browser-based password manager. You can register an account, securely store credentials for any website or service, and access them from any device with a web browser. No installation required.

Access the application at: <https://vault.neuralnetworks.me/>

Key features:

- Encrypted vault for storing usernames, passwords, and websites
- Built-in password generator
- Two-factor authentication (MFA) for account security
- Search and category filtering
- All credentials are encrypted in your browser before being sent to our servers, we never see your plaintext passwords

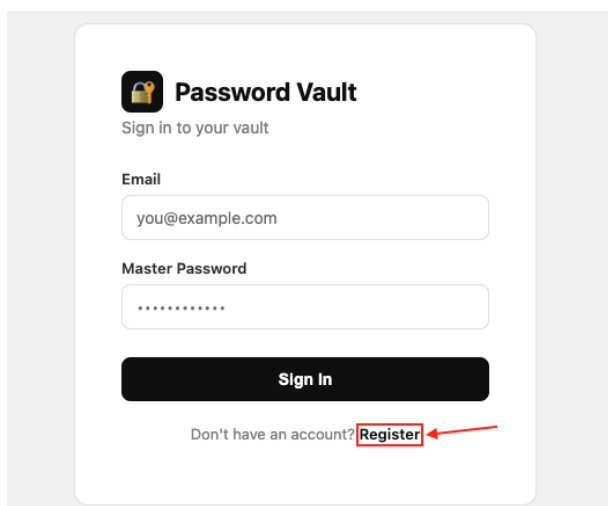
What you'll need:

- A modern web browser (Chrome, Safari, Firefox, or Edge)
- An authenticator app on your phone (Google Authenticator, Authy, 1Password, Microsoft Authenticator, or any TOTP-compatible app)

1. Creating Your Account

1.1 Open <https://vault.neuralnetworks.me/> in your browser. You'll land on the Sign In page.

1.2 Click "Register" at the bottom of the sign-in card to go to the registration page.

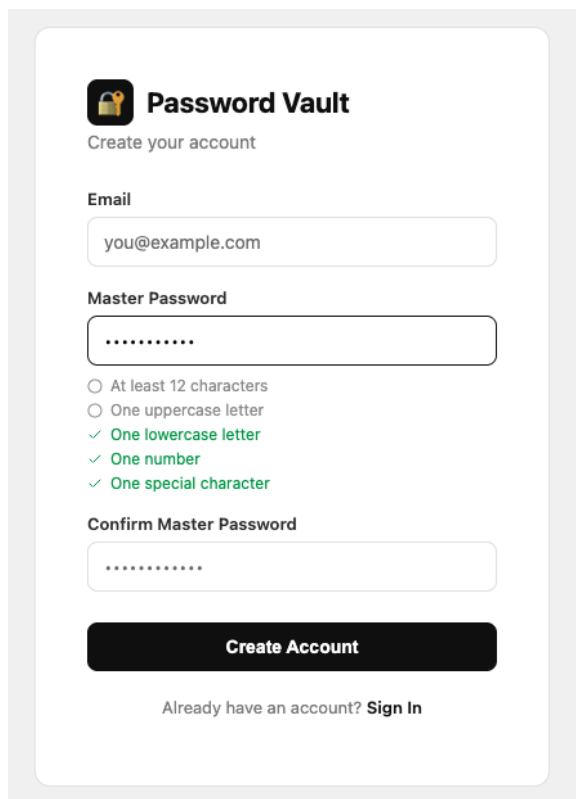


1.3 Enter your email address.

1.4 Create your Master Password. This is the most important password you'll ever set. It protects every credential you save in your vault. Your master password must meet these requirements:

- At least 12 characters
- One uppercase letter (A–Z)
- One lowercase letter (a–z)
- One number (0–9)
- One special character (! @ # \$ % ^ & * etc.)

As you type, you'll see real-time checkmarks next to each requirement so you know when your password is strong enough.



The screenshot shows a web form for creating a Password Vault account. At the top is the 'Password Vault' logo and the text 'Create your account'. Below this is an 'Email' field containing 'you@example.com'. The 'Master Password' field is shown with dots, followed by a list of requirements: 'At least 12 characters' (unchecked), 'One uppercase letter' (unchecked), 'One lowercase letter' (checked with a green checkmark), 'One number' (checked with a green checkmark), and 'One special character' (checked with a green checkmark). Below the requirements is a 'Confirm Master Password' field, also with dots. At the bottom is a large black 'Create Account' button. Below the button is a link that says 'Already have an account? Sign In'.

1.5 Re-enter your Master Password in the "Confirm Master Password" field. Both passwords must match.

1.6 Click "Create Account".

IMPORTANT: Remember your master password. We cannot reset it for you. If you forget it, your vault and all credentials inside it become permanently unrecoverable.

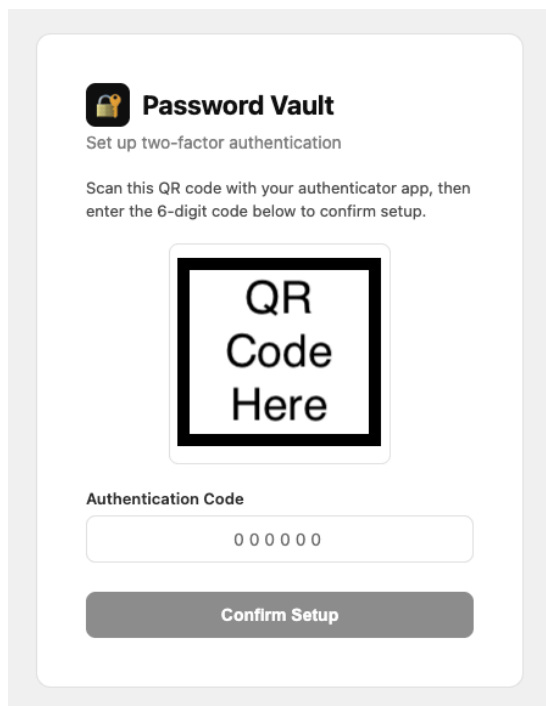
2. Setting Up Two-Factor Authentication (MFA)

After registering, you'll be taken directly to the MFA setup page. MFA adds a second layer of security to your account. Even if someone learns your master password, they can't access your vault without the 6-digit code from your phone.

2.1 Open your authenticator app on your phone.

2.2 In your authenticator app, choose "Add account" (or "Scan QR code" - the exact label depends on which app you use).

2.3 Scan the QR code shown on the screen with your phone's camera.



2.4 Your authenticator app will add "SecurePasswordManager" as a new entry and immediately start showing a 6-digit code that changes every 30 seconds.

2.5 Type the current 6-digit code into the "Authentication Code" field on the setup page.

2.6 Click "Confirm Setup".

You'll be taken straight into your vault. MFA is now active on your account, and you'll be prompted for a code from your authenticator app every time you log in.

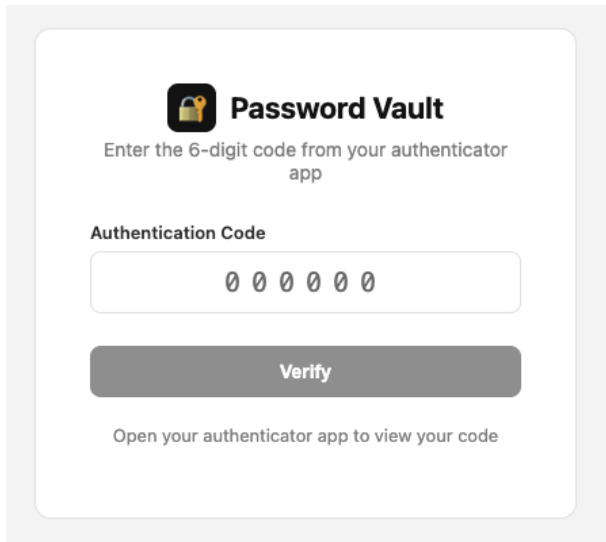
3. Logging In

3.1 Go to <https://vault.neuralnetworks.me/>.

3.2 Enter your email and master password.

3.3 Click "Sign In".

3.4 You'll be prompted for your 6-digit MFA code. Open your authenticator app, find the "SecurePasswordManager" entry, and type in the current code.



Password Vault

Enter the 6-digit code from your authenticator app

Authentication Code

0 0 0 0 0 0

Verify

Open your authenticator app to view your code

3.5 Click "Verify". You're now in your vault.

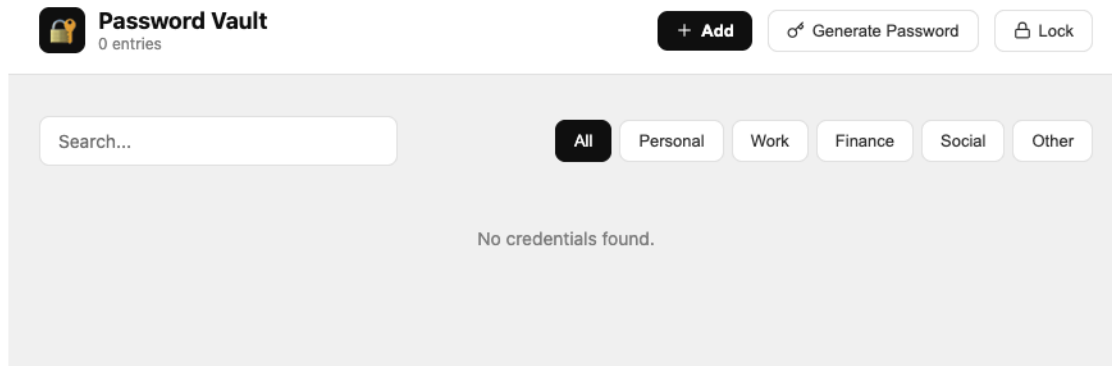
4. Your Vault

After logging in, you'll see your vault. The page has:

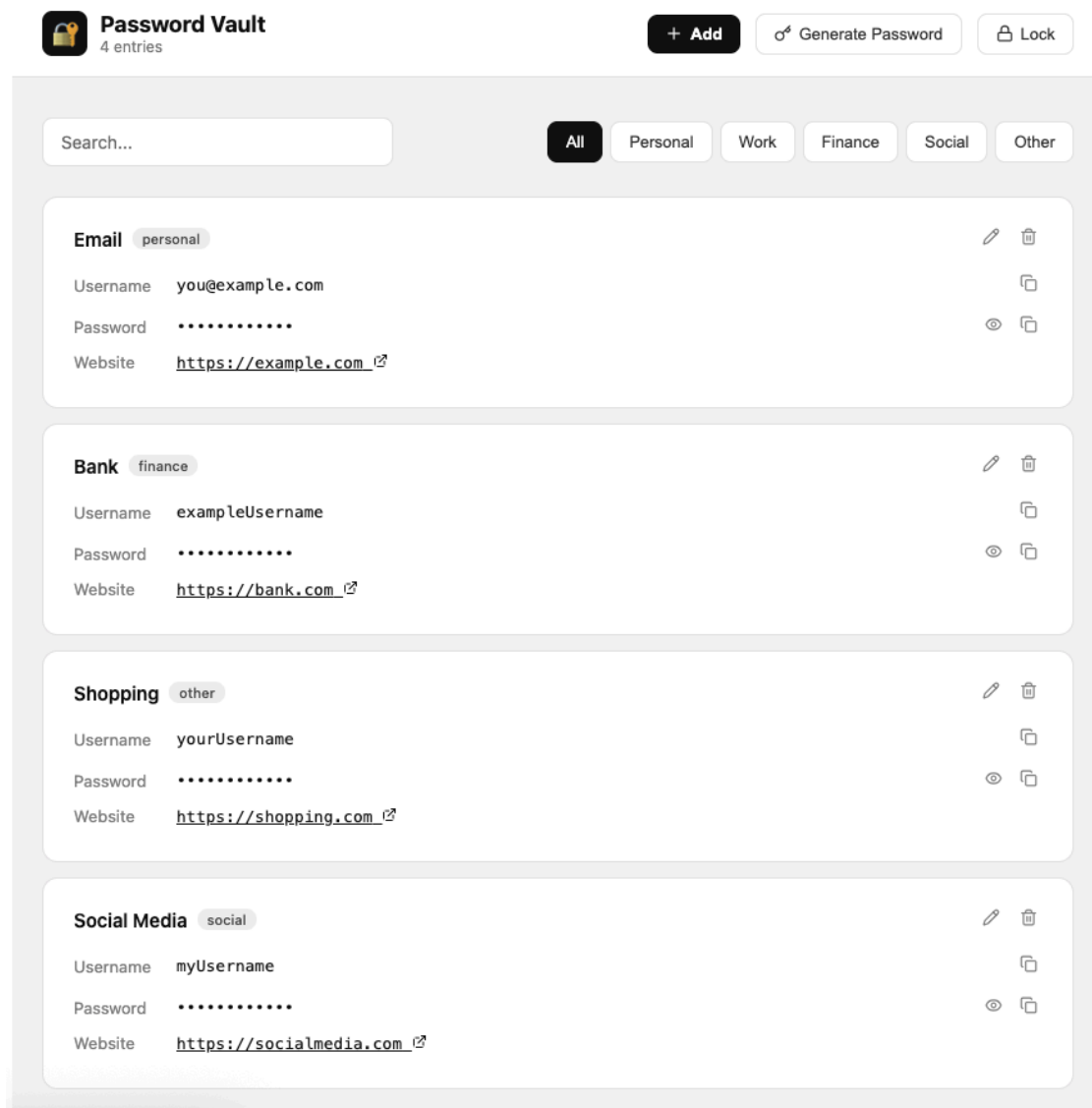
- A header with your entry count and three action buttons (Add, Generate Password, Lock)
- A search bar
- Category filter buttons (All, Personal, Work, Finance, Social, Other)
- A list of your saved credentials, each shown as a card

If this is a new account, your vault will be empty.

Empty Vault:



Vault with Credentials Saved:

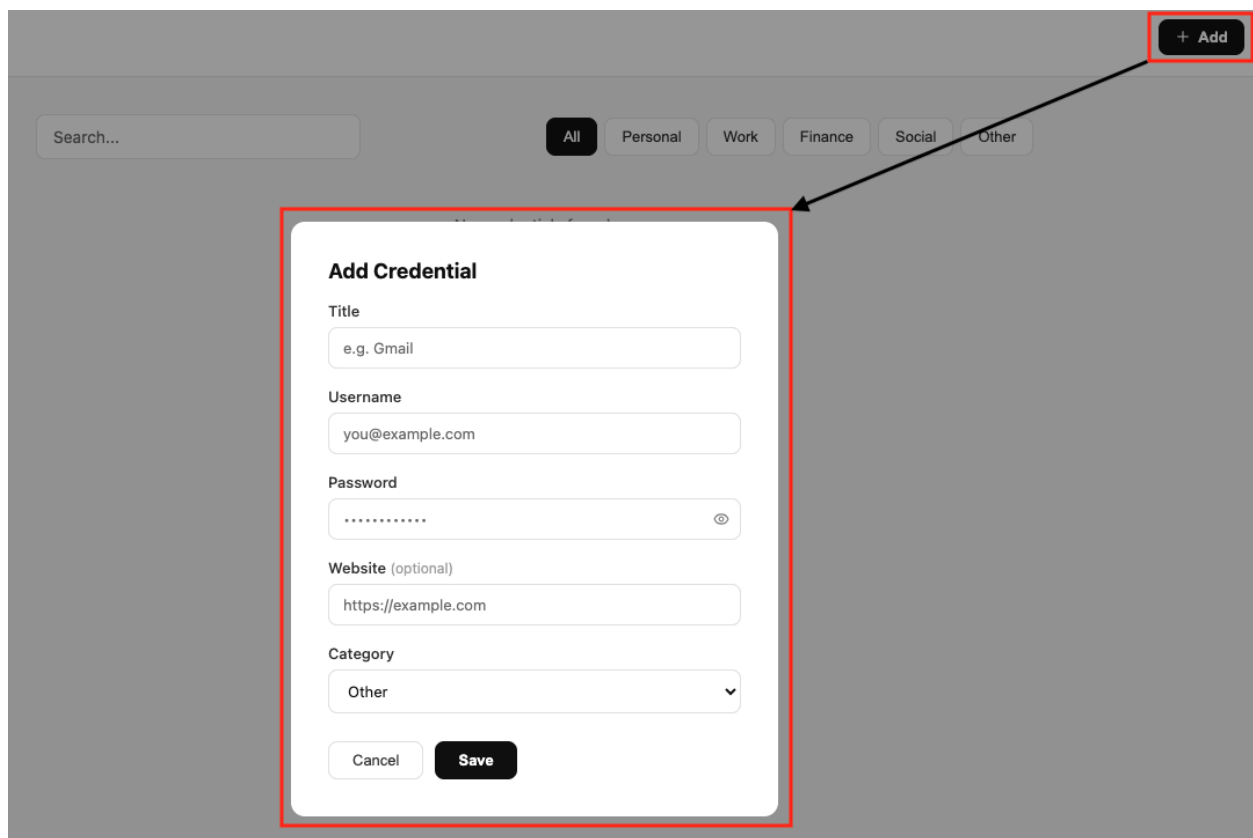


5. Adding a Credential

5.1 Click the "+ Add" button in the top right of the vault.

5.2 A form will open. Fill in:

- **Title** (required): A name to recognize this entry. Example: "Gmail" or "Bank of America".
- **Username** (required): Your username, email, or account ID for this service.
- **Password** (required): The password for this service. Click the eye icon to show/hide text.
- **Website** (optional): The URL for this service. Example: "https://gmail.com".
- **Category**: Pick one: Personal, Work, Finance, Social, or Other.



The screenshot shows a web application interface for a password manager. At the top right, there is a dark button with a white plus sign and the text "+ Add", which is highlighted with a red rectangle. Below this, a modal form titled "Add Credential" is displayed, also outlined with a red rectangle. The form contains the following fields: "Title" with the placeholder "e.g. Gmail", "Username" with the placeholder "you@example.com", "Password" with a masked input (dots) and an eye icon, "Website (optional)" with the placeholder "https://example.com", and a "Category" dropdown menu currently set to "Other". At the bottom of the form are "Cancel" and "Save" buttons. In the background, a search bar and category tabs ("All", "Personal", "Work", "Finance", "Social", "Other") are visible.

5.3 Click "Save". The credential is encrypted in your browser and saved to your vault. You'll see a "Credential saved." confirmation at the bottom of the screen.

To cancel without saving, click "Cancel".

6. Viewing and Using Saved Credentials

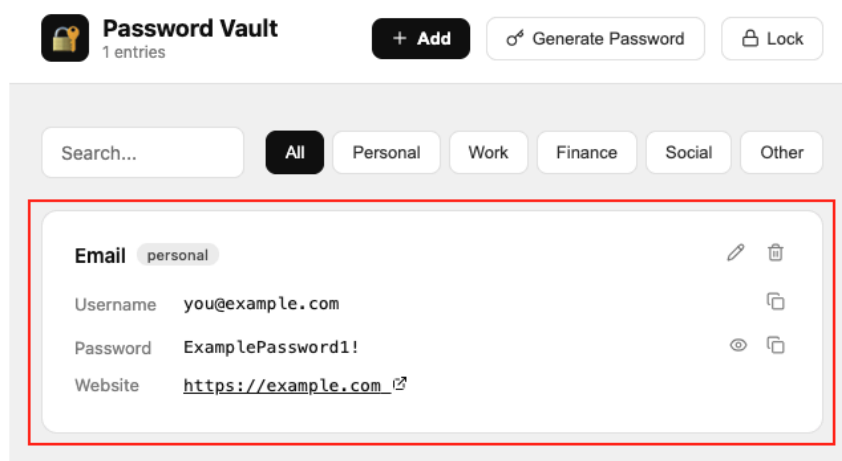
Each credential card shows the title, category, username, and a hidden password (••••••••). You can:

6.1 Show the password — Click the eye icon next to the password field to reveal it. Click again to hide it.

6.2 Copy the username — Click the copy icon next to the username. The icon will change to a checkmark for 2 seconds to confirm. You can now paste your username into another website.

6.3 Copy the password — Click the copy icon next to the password. Same confirmation behavior.

6.4 Open the website — If you saved a website URL, click it on the card to open the site in a new tab.



7. Searching and Filtering

7.1 Search — Type into the search bar at the top. The list filters in real time, matching either the title or the username of your credentials.

7.2 Filter by category — Click any of the category buttons (All, Personal, Work, Finance, Social, Other) to show only credentials in that category. "All" shows everything.

You can combine search and filter. For example, filter to "Finance" and then search for "bank" to find only banking entries.

Credentials Filtered by Category:

Password Vault
3 entries

+ Add Generate Password Lock

Search...

All Personal Work **Finance** Social Other

Stock Portfolio finance

Username exampleUsername

Password

Website <https://stocks.com>

Bank finance

Username you@example.com

Password

Website <https://yourbank.com>

Credentials Filtered by Category and Searched Key Word:

Password Vault
3 entries

+ Add Generate Password Lock

Bank Search...

All Personal Work **Finance** Social Other

Bank finance

Username you@example.com

Password

Website <https://yourbank.com>

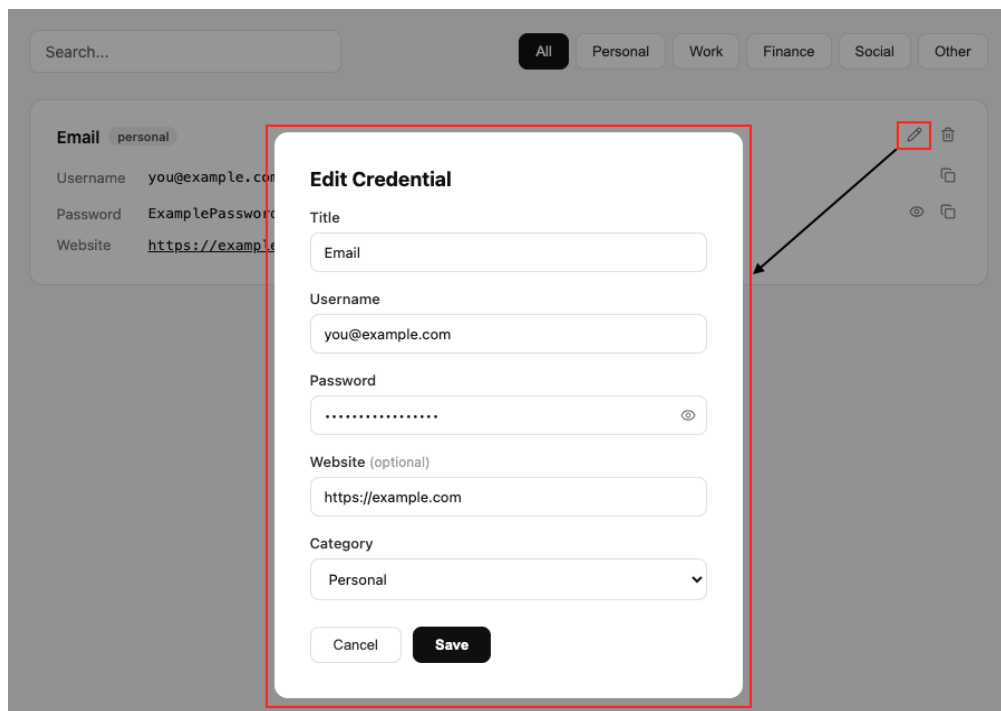
8. Editing a Credential

8.1 Find the credential you want to change.

8.2 Click the pencil icon in the top right of the card.

8.3 An edit form opens with the current values pre-filled. Modify any field — title, username, password, website, or category.

8.4 Click "Save" to apply your changes, or "Cancel" to discard them.



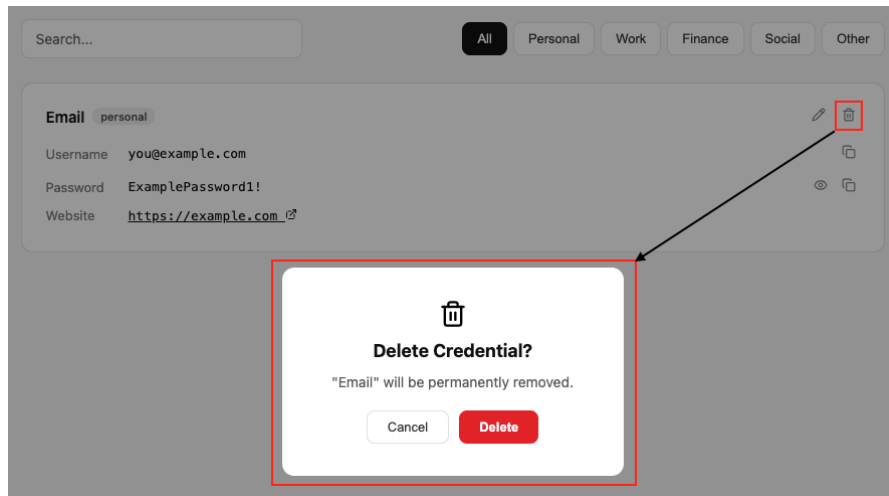
9. Deleting a Credential

9.1 Click the trash icon in the top right of the credential card.

9.2 A confirmation dialog appears asking if you're sure. The credential's title is shown in the message so you can double-check.

9.3 Click "Delete" to permanently remove the credential, or "Cancel" to keep it.

WARNING: Deleted credentials cannot be recovered. Make sure you've copied any information you need before deleting.

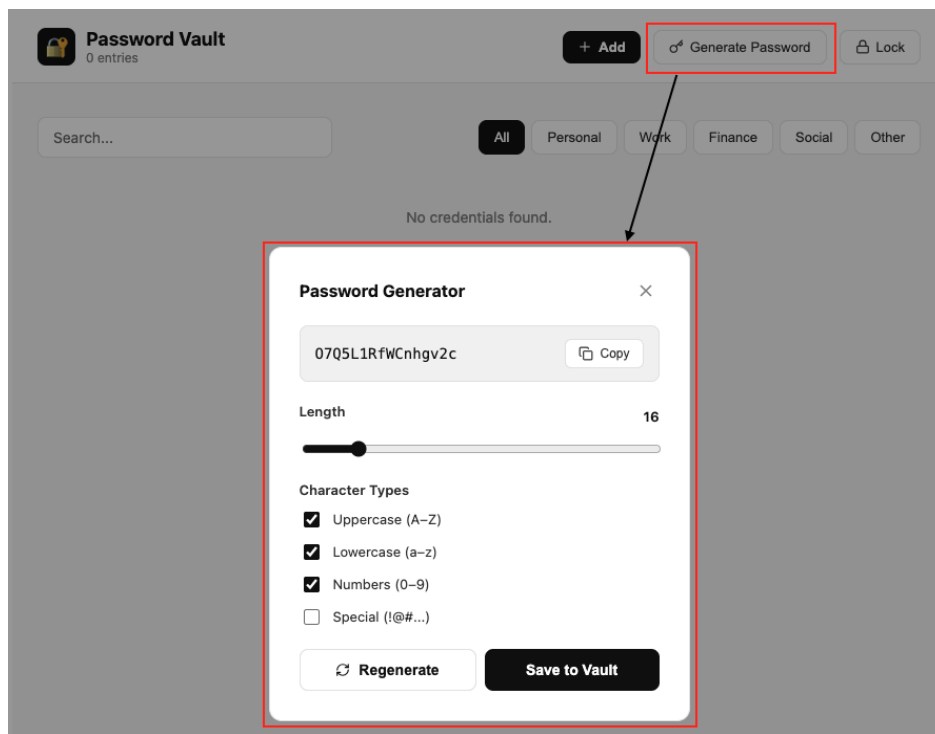


10. Using the Password Generator

The built-in password generator creates strong, random passwords you can use for any new account.

10.1 Click "Generate Password" in the top right of the vault.

10.2 A generator window opens with a freshly generated password.



10.3 Adjust the password to your needs:

- **Length:** Drag the slider to set the password length anywhere from 8 to 64 characters.
- **Character Types:** Toggle which kinds of characters to include:
 - Uppercase (A–Z)
 - Lowercase (a–z)
 - Numbers (0–9)
 - Special (! @ # \$ % ^ & * etc.)

At least one type must stay enabled.

10.4 Click "Regenerate" to create a new password with your current settings. Repeat until you get one you like.

10.5 When you have a password you want to use:

- Click "Copy" to copy it to your clipboard. Paste it wherever you're creating an account.
- Click "Save to Vault" to open the Add Credential form with the generated password already filled in. Then fill in the title, username, website, and category, and save.

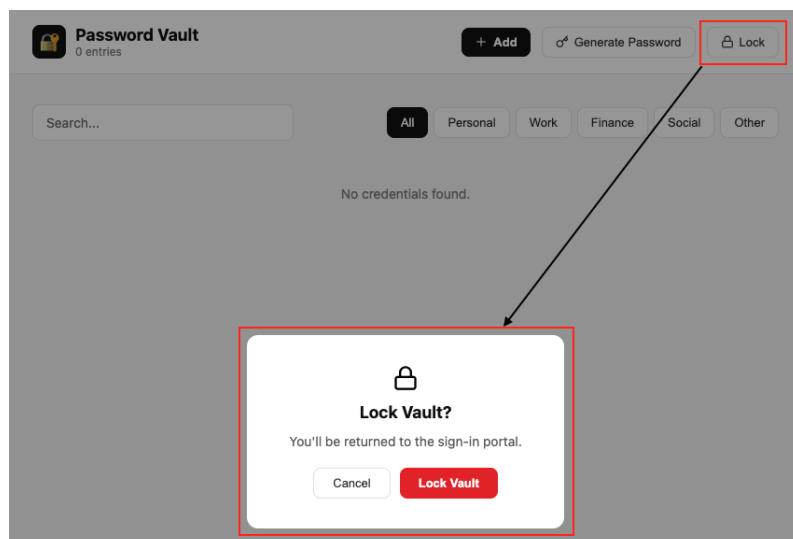
10.6 To close the generator without using the password, click the X in the corner.

11. Locking Your Vault

When you're done using the vault, lock it. Locking signs you out — you'll need to log back in with your email, master password, and MFA code to access it again.

11.1 Click "Lock" in the top right.

11.2 A confirmation dialog appears. Click "Lock Vault" to confirm, or "Cancel" to stay signed in.

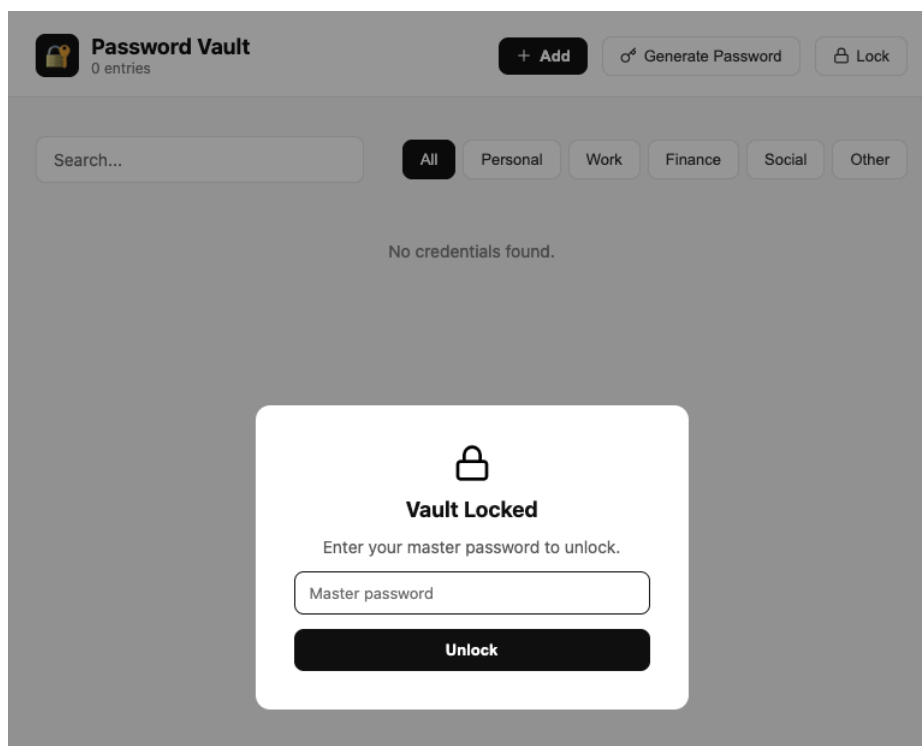


After locking, you'll be returned to the Sign In page.

TIP: Always lock your vault when you step away from your computer, especially on shared or public devices.

12. Reloading the Page

If you reload the vault page (or close and reopen the tab) while still signed in, you'll see an "Unlock Vault" prompt instead of being signed out completely.



This happens because your master password is never stored anywhere. It only lives in your browser's active memory while you're using the vault. When you reload, the memory clears.

To continue:

12.1 Type your master password into the prompt.

12.2 Press Enter or click "Unlock". Your credentials decrypt and the vault loads.

If you enter the wrong master password, you'll see "Incorrect master password. Try again." Type it again carefully.

13. Security Notes

- Your master password is never sent to our servers. Only a one-way hash is stored for verifying your identity at login.
- Every credential you save is encrypted with AES-256 in your browser before it's sent. Our database only ever holds encrypted ciphertext. We cannot read your passwords even if we wanted to.
- Two-factor authentication is required on every login. Even if someone steals your master password, they can't get in without your phone.
- All traffic between your browser and our servers is encrypted over HTTPS.
- If you suspect your master password has been compromised, log in immediately and consider creating a new account, since the master password cannot be changed in the current version of the app.

14. Troubleshooting

"Invalid credentials" when signing in

Double-check your email and master password. After 5 failed attempts in a row, your account locks for 15 minutes. Wait it out and try again.

"Invalid MFA token"

The 6-digit code expired (codes refresh every 30 seconds). Wait for a new code in your authenticator app and try again. Make sure your phone's clock is accurate.

"Incorrect master password" after reload

Make sure caps lock is off and you're typing the exact master password you registered with. If you've forgotten it, the vault is unrecoverable. You'll need to register a new account.

Lost your phone or authenticator app

Account recovery without your MFA device is not supported in the current version. Keep backup access to your authenticator (most apps offer cloud sync - Authy, Microsoft Authenticator, 1Password) or save the secret in a secure place when you set up MFA.

The page won't load or shows an error

Try a hard refresh (Cmd+Shift+R on Mac, Ctrl+Shift+R on Windows). If the problem persists, check that you're using a modern, up-to-date browser.

SUPPORT

This is an open-source CS 467 capstone project at Oregon State University. The full source code is available on GitHub, and the application is provided as-is for educational and personal use.